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## NATIONAL UNIVERSITY OF SINGAPORE

### FINAL ASSESMENT FOR CS2105 – INTRODUCTION TO COMPUTER NETWORKS

(Semester 1: AY2022/2023)

Time allowed: 120 minutes

#### INSTRUCTIONS TO CANDIDATES

1. This assessment paper comprises 15 printed pages, including this page.
2. The total maximum score is **50 marks**.
3. Answer all the questions using the knowledge you learned in CS2105.
4. This is an **OPEN BOOK** assessment. You may bring in any printed materials. Calculators are also allowed, but no laptops or other electronic devices.
5. Each MCQ has one correct answer. Answer each MCQ by shading the letter corresponding to the correct answer on the **OAS form** provided. Shade and write down your **student number** on the OAS form too. You must use a **2B** pencil to shade on the OAS form, or the grading machine might not be able to register your shading.
6. **Submit** both the OAS form and the question paper at the end of the assessment.
7. Leave your student card on the desk throughout this assessment.
8. Do **NOT** look at the questions until you are told to do so.
9. Fill in your student number clearly on all pages.
10. Please write your Student Number only. Do not write your name.

This portion is for examiner's use only

| Question | Marks | Remarks |
|----------|-------|---------|
| Q25      |       |         |
| Q26      |       |         |
| Q27      |       |         |
| Q28      |       |         |
| Q29      |       |         |
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CS2105

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## Part 3

### Short Answer Questions (16 Marks)

For each of the question below, kindly enter your answers in the space provided. Each question is worth 1 Mark.

25. [1 mark] A sender and a receiver are directly connected by a cable 10 kilometers long. The link bandwidth is 1 Mbps, and the propagation speed of the signal in the link is  $2 \times 10^8$  meters/second. The sender sends a 40-byte probe packet to the receiver, and the latter immediately replies with a 20-byte acknowledgement packet.

How long (in milliseconds) will it take for this ACK to reach the sender (from the time the sender begins its transmission of the probe packet)?

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26. [3 marks] Consider a protocol that computes 1's complement checksum over 8-bit data.

- (a) Suppose the receiver receives a packet containing the following three bytes: **10001010**, **11001100** and **10101010**. The first two bytes are data, and the last byte is the checksum. Will the packet pass checksum verification? Answer "yes" or "no".

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- (b) Suppose the receiver receives another packet containing the following three bytes: **10001010**, **11001100** and **10101000**. One of the three bytes is the checksum, but the receiver doesn't know which byte it is. Will the packet pass checksum verification? Answer "yes" or "no".

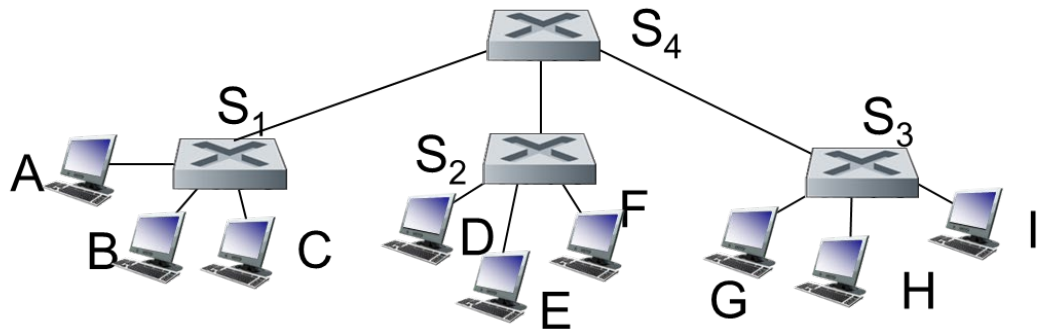
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- (c) In no more than 50 words, justify your answer in (b) above.

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27. [5 marks] Consider the following topology of switched Ethernet. In which no node has sent out any packets. (Here, nodes refer to machines other than Switches)



- (a) Node A sends an IP datagram addressed to destination 255. 255. 255. 255. What is the destination MAC address of the corresponding Ethernet frame?

- (b) How many nodes will receive the frame sent in (a)?

- (c) At the end of (a), what is the ARP table entry of  $S_1$ ? You may assume a TTL value of 60, and you may represent the IP address and MAC address of a node  $x$  to be  $IP_x$  and  $MAC_x$  respectively.

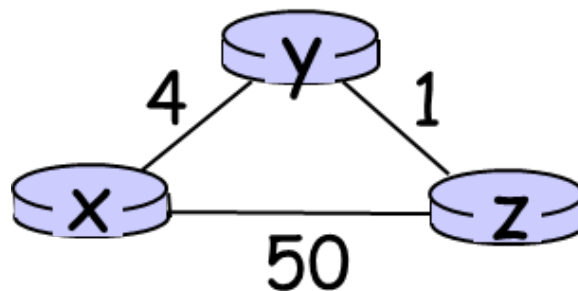
- (d) After some time of the network operation, all pairs of nodes have communicated with each other. Now, we add a cable connecting switch  $S_1$  to  $S_2$  (the rest of the topology is unchanged). Though we have been advised to avoid loops in the network topology, this will not cause any problems since the switch tables are fully populated and stable. Answer "yes" or "no".

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(e) In no more than 50 words, justify your answer in (d) above.

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28. [5 marks] Consider the following topology of interconnected routers running the Distance Vector Algorithm. The routing tables in all the routers have converged.



(a) What is the distance of  $x$  from  $z$ ,  $d_z(x)$ ?

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(b) Due to some changes in network policy, we changed the cost of the link  $(x, y)$  to 60. Router  $y$  has updated its routing tables (It has not yet exchanged its local view with any other router). What is the value of  $d_y(x)$ ?

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(c) After the events of (b) above, router  $y$  has exchanged the local view with  $z$ . Subsequently,  $z$  updates its local view. What is the value of  $d_z(x)$ ?

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(d) After the routing tables have converged, what is the value of  $d_z(x)$ ?

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- (e) If we define an iteration to be the mutual exchange of local views between  $y$  and  $z$  and the related update of respective local views; how many iterations does it take for the routing tables to converge after the events of (b) above?

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29. [2 marks] Consider RSA with  $p = 7$  and  $q = 13$ .

- (a) Is the choice of 17 a valid choice for  $e$ ? Answer “yes” or “no”.

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- (b) What is the private key,  $K^-$ ?

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**=== End of Paper ===**